Blackjack By Bender!

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Let’s go gambling!

Overview

- Objective
- Requirements
- Implementation
  - Basic Game Loop
  - Testing
- Reflections
- Demo

Defining Requirements

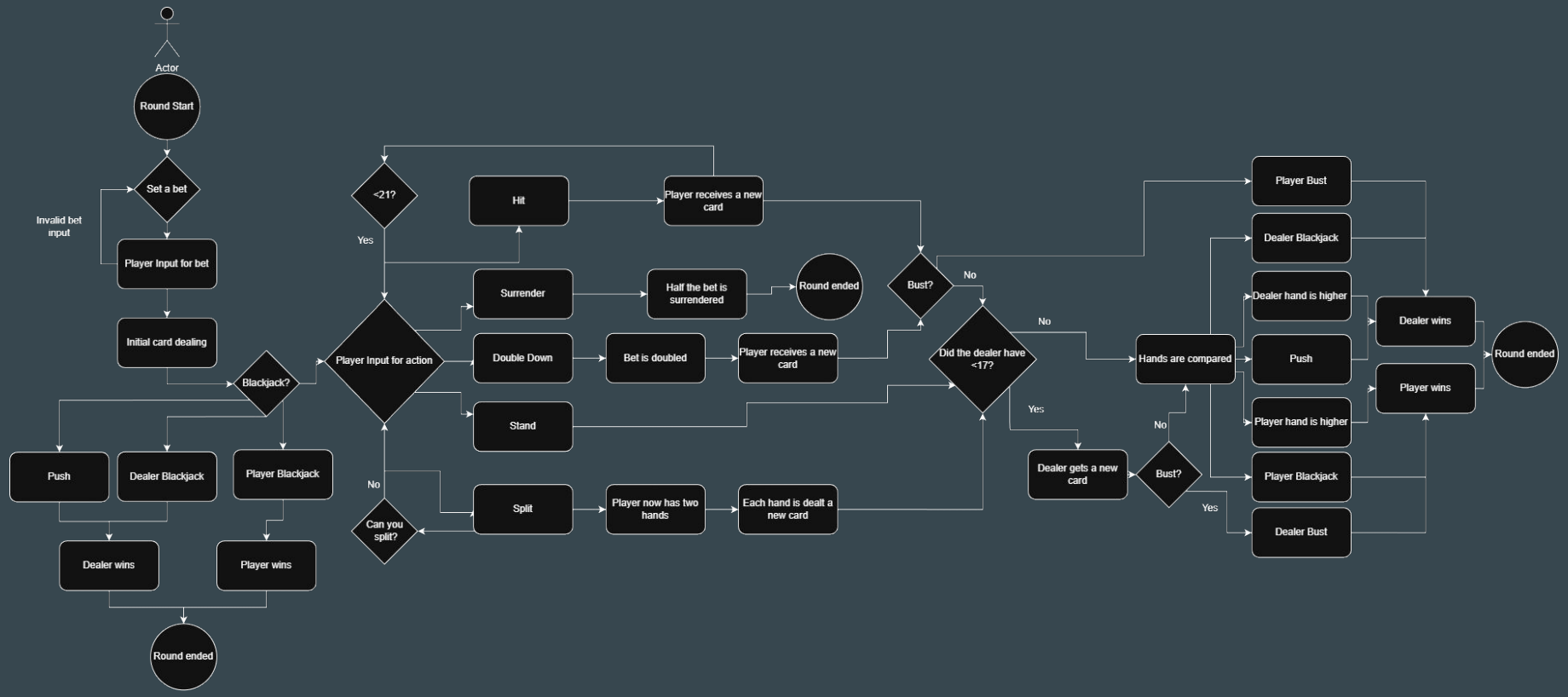
MVP (Essential)

- End states (Bust, Blackjack, Push)
- All cards have a numerical value
- Value of Ace is conditional
- Cards are unique
- Player is able to write an input to the console (Hit/Stand)

Nice-To-Have

- Splitting is an option
- The player has a number of chips, with which they can bet, using console input
- The player can Double Down or Surrender as an optional input

Implementation - Basic Game Loop



Implementation - Testing

Unit Testing

- NUnit
- Simulated inputs
- Test cases
  - Complete all inputs
  - Push
  - Blackjack detection
  - Invalid inputs
- Deterministic cards

```
[Test]
0 references
public void TestSplitInitialDeal_Deterministic()
{
    // Arrange: player dealt two 8s so split is possible
    var bet = new Bet { CurrentChips = 1000 };
    var cards = new System.Collections.Generic.List<Card>
    {
        new Card("Hearts", "8"),    // player 1
        new Card("Spades", "8"),    // player 2
        new Card("Clubs", "10"),    // dealer 1
        new Card("Diamonds", "9"),  // dealer 2
        new Card("Hearts", "5"),    // split new card for hand1
        new Card("Spades", "6")     // split new card for hand2
    };
    var gen = new TestCardGenerator(cards);
    var game = new BlackjackGame(gen, bet);
    SimulateUserInput("100\np\n");

    // Act
    game.PlayRound();

    // Assert: after split, the game's dealer should have two split hands with two cards each
    Assert.That(game.dealer.playerHand1.Count, Is.EqualTo(2), "Player's first split hand should have 2 cards.
    Assert.That(game.dealer.playerHand2.Count, Is.EqualTo(2), "Player's second split hand should have 2 cards
}
```

Reflections

Challenges / Pitfalls

- Splitting
  - Hand separation
  - New end states
- Bug fixing
  - Infinite rounds
  - Blackjack detection
- Unit testing
  - New experience

Future works

- More test cases
  - Test more methods
  - Find more edge cases
- Strings vs Enums
  - Type safety
  - Performance
- OOP readability

DEMO

Who wants to go gambling?